API Implementation Documentation - Animal Adoption System

GET Routes

Functionalities of Web Application	Related SQL Code (Full and complete SQL Statements, not partial)	URL Route Pattern	Payload	HTTP Method	Result From Postman (Pass/Fail)
Get all animals with their default images and species	`SELECT animal.*, animal_images.image_url, animal_species.species_name FROM animal LEFT JOIN animal_images ON animal_images.animal_id = animal.animal_id AND animal_images.is_default = 1 LEFT JOIN animal_species ON animal.species_id = animal_species.species_id`	`/animal`	N.A.	GET	(To be filled by Lecturer)
Get specific animal details by animal ID with default image and species	`SELECT animal.*, animal_images.image_url, animal_species.species_name FROM animal LEFT JOIN animal_images ON animal_images.animal_id = animal.animal_id AND animal_images.is_default = 1 LEFT JOIN animal_species ON animal.species_id = animal_species.species_id WHERE animal.animal_id = ?`	`/animal/:animal_id`	N.A.	GET	(To be filled by Lecturer)
Get all animal images (for testing/debugging)	`SELECT * FROM animal_images`	`/animal_image`	N.A.	GET	(To be filled by Lecturer)
Get specific image by image ID	`SELECT * FROM animal_images WHERE image_id = ?`	`/animal_image/:image_id`	N.A.	GET	(To be filled by Lecturer)

Functionalities of Web Application	Related SQL Code (Full and complete SQL Statements, not partial)	URL Route Pattern	Payload	HTTP Method	Result From Postman (Pass/Fail)
Get all images for a specific animal	SELECT * FROM animal_images WHERE animal_id = ?	/animal/:animal_id/images	N.A.	GET	(To be filled by Lecturer)
Get all species (for testing/debugging)	SELECT species_id, species_name FROM animal_species ORDER BY species_name	/species	N.A.	GET	(To be filled by Lecturer)

POST Routes

Functionalities of Web Application	Related SQL Code (Full and complete SQL Statements, not partial)	URL Route Pattern	Payload	HTTP Method	Result From Postman (Pass/Fail)
Add a new animal (without images)	INSERT INTO animal_adoption.animal (name, age_year, age_month, species_id, breed, gender, temperament, adoption_status) VALUES (?, ?, ?, ?, ?, ?, ?, ?)	/animal	{ "name": "string", "age_year": number, "age_month": number, "species_id": number, "breed": "string", "gender": "string", "temperament": "string", "adoption_status": "string" }	POST	(To be filled by Lecturer)
Add an animal image	INSERT INTO animal_adoption.animal_images (animal_id, image_url, is_default) VALUES (?, ?, ?)	/animal_image	{ "animal_id": number, "image_url": "string", "is_default": number }	POST	(To be filled by Lecturer)

PUT Routes

Functionalities of Web Application	Related SQL Code (Full and complete SQL Statements, not partial)	URL Route Pattern	Payload	HTTP Method
Update animal information (without images)	`UPDATE animal_adoption.animal SET name=?, age_year=?, age_month=?, species_id=?, breed=?, gender=?, temperament=?, adoption_status=? WHERE animal_id=?`	`/animal/:animal_id`	`{ "name": "string", "age_year": number, "age_month": number, "species_id": number, "breed": "string", "gender": "string", "temperament": "string", "adoption_status": "string", "animal_id": number }`	PUT
Update default image for an animal	`UPDATE animal_adoption.animal_images SET image_url=? WHERE animal_id=? AND is_default=1`	`/animal_image/default/:animal_id`	`{ "image_url": "string" }`	PUT
Update a specific animal image	`UPDATE animal_adoption.animal_images SET animal_id=?, image_url=? WHERE image_id=?`	`/animal_image/:image_id`	`{ "animal_id": number, "image_url": "string" }`	PUT
Update only the adoption status of an animal	`UPDATE animal_adoption.animal SET adoption_status = ? WHERE animal_id = ?`	`/animalstatus/:animal_id`	`{ "adoption_status": "string" }`	PUT

DELETE Routes

Functionalities of Web Application	Related SQL Code (Full and complete SQL Statements, not partial)	URL Route Pattern	Payload	HTTP Method	Result From Postman (Pass/Fail)
Delete a specific animal image	`DELETE FROM animal_adoption.animal_images WHERE image_id=?`	`/animal_image/:image_id`	N.A.	DELETE	(To be filled by Lecturer)
Delete an animal (cascades to delete related images)	`DELETE FROM animal_adoption.animal WHERE animal_id = ?`	`/animal/:animal_id`	N.A.	DELETE	(To be filled by Lecturer)

Notes:

- All routes use `http://127.0.0.1:8084` as the base URL
- Parameters in the URL are denoted with `:` (e.g., `:animal_id`)
- Payload examples show the JSON structure expected in the request body
- N.A. means no payload is required for that route
- Question marks (?) in SQL represent parameterized values to prevent SQL injection